

House Price Prediction using Linear Regression

AIML Internship – Task 1

Submitted by: Rithvik A

Date: 21/02/2026

ABSTRACT

This project focuses on predicting house prices using a supervised machine learning algorithm, Linear Regression. The California Housing dataset from scikit-learn was used for model training and evaluation. The objective was to understand the end-to-end machine learning workflow, including data preprocessing, exploratory data analysis, model building, and performance evaluation.

The developed model achieved an R^2 score of **0.5758**, demonstrating moderate predictive capability. This project provided practical exposure to regression modeling and performance evaluation techniques.

1. INTRODUCTION

House price prediction is a classic regression problem in machine learning. Accurate prediction models help in real estate valuation, investment planning, and financial analysis.

In this project, a Linear Regression model was developed to estimate median house values based on socio-economic and geographical features from the California Housing dataset.

The primary goals were:

- To perform Exploratory Data Analysis (EDA)
- To build a regression model
- To evaluate model performance using standard metrics
- To interpret results and suggest improvements

2. DATASET DESCRIPTION

The dataset used is the **California Housing dataset**, available within the scikit-learn library.

Dataset Details:

- **Total Samples:** 20,640
- **Total Features:** 8 numerical features
- **Target Variable:** Median House Value (MedHouseVal)

Features:

- Median Income (MedInc)
- House Age (HouseAge)
- Average Rooms (AveRooms)
- Average Bedrooms (AveBedrms)
- Population
- Average Occupancy (AveOccup)
- Latitude
- Longitude

The dataset contained **no missing values**, which simplified preprocessing.

3. EXPLORATORY DATA ANALYSIS (EDA)

EDA was performed to understand data distribution and relationships.

Steps Performed:

- Data inspection using `df.info()` and `df.describe()`
- Checking for null values
- Correlation matrix visualization using heatmap

Key Observations:

- Median Income showed the strongest positive correlation with house prices.
- Location-based features (Latitude and Longitude) significantly influence prices.
- No missing or invalid entries were found.

The correlation heatmap helped identify feature importance and relationships among variables.

4. METHODOLOGY

4.1 Data Splitting

The dataset was divided into:

- 80% Training Data
- 20% Testing Data

This ensured unbiased evaluation of the model's predictive performance.

4.2 Model Implementation

The Linear Regression model was implemented using scikit-learn.

Linear Regression models the relationship between dependent and independent variables using a linear equation:

$$y = b_0 + b_1x_1 + b_2x_2 + \dots + b_nx_n$$

Where:

- y = Target variable
- x = Input features
- b = Coefficients

The model was trained on the training dataset and evaluated on unseen test data.

5. MODEL EVALUATION

The performance of the model was evaluated using:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- R^2 Score

Results:

- **MAE:** 0.5332
- **MSE:** 0.5559

- **R² Score:** 0.5758

Interpretation:

- The MAE value indicates that the average prediction error is approximately 0.53 units.
- The MSE shows moderate error variance.
- The R² score of 0.5758 indicates that the model explains nearly **57.58% of the variance** in house prices.

This level of accuracy is reasonable for a basic Linear Regression model without feature engineering or hyperparameter tuning.

6. VISUALIZATIONS

The following visualizations were generated:

1. Correlation Heatmap

Displayed relationships between all numerical features and the target variable.

2. Actual vs Predicted Plot

Compared real house prices with predicted values.

The scatter distribution followed a near-diagonal pattern, indicating moderate predictive accuracy.

(Screenshot images should be inserted here in the final PDF.)

7. CONCLUSION

This project successfully implemented a complete machine learning workflow for house price prediction.

Key achievements:

- Data exploration and preprocessing
- Regression model training
- Performance evaluation using multiple metrics
- Result interpretation

The Linear Regression model demonstrated moderate predictive capability. While not highly accurate, it establishes a strong baseline for further model enhancement.